

Emergent Tipping Points in AI Labour Markets: Design and Outcomes of a Task-Based Agent-Based Simulator

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Abstract

How artificial intelligence reshapes labour markets depends on interactions between firm-level adoption decisions and worker-level reallocation that aggregate models struggle to represent without assuming their conclusions. This paper describes the design of an open, task-based agent-based model (ABM) in which roughly 5,000 workers and 400 firms interact monthly: firms adopt AI task-by-task when quality-adjusted AI costs undercut wages, amplified by sectoral imitation and heterogeneous organizational hurdles, while displaced workers search, bargain, and retrain across an occupation space whose geometry derives from task-content similarity. No functional form for diffusion is imposed; adoption S-curves, displacement waves, wage scarring, and Beveridge-curve loops all emerge from micro-interactions. A no-AI baseline is calibrated to empirical labour-flow anchors and locked as an automated test gate before any AI mechanism is introduced. Simulation experiments show that a fast-takeoff scenario raises output by roughly 80% while labour-market distress climbs from an early displacement spike near 20% of the workforce toward a third under sustained automation, that median wages halve while within-labour dispersion *compresses* toward the benefit floor—shifting the surplus toward firms rather than spreading it among workers—and that canonical policy levers are strikingly non-robust: unemployment-benefit generosity lowers median peak distress in the final specification yet worsens a minority of seeds and reversed sign under an earlier revision of the demand block; employment protection leaves median peak distress essentially unchanged while flipping sign seed-by-seed; only the efficiency of reallocation itself delivers a consistent median improvement. The model ships as an interactive web application with Monte Carlo uncertainty quantification and full seed-level reproducibility. We close with an agenda for empirical curation, mechanism extensions, and validation.

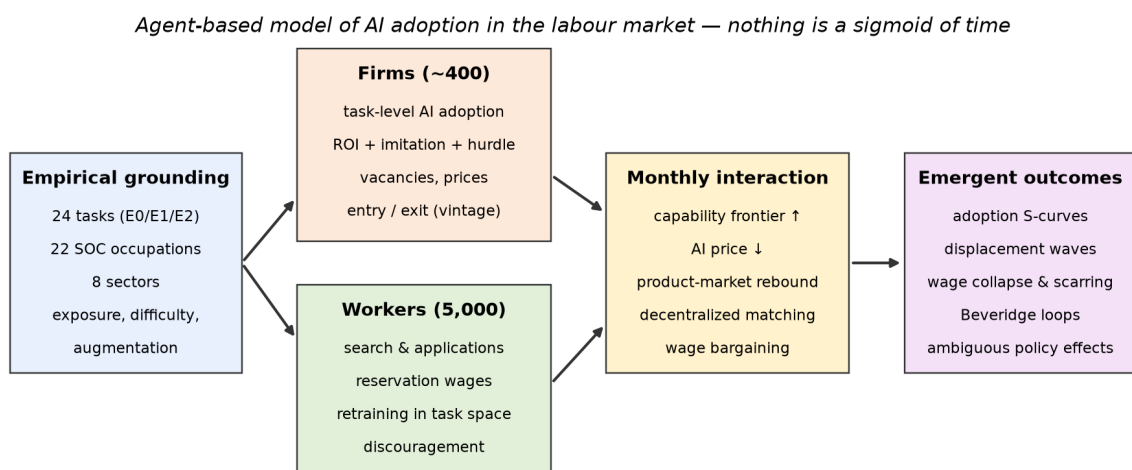


Figure 1: Graphical abstract. Empirically grounded tasks, occupations, and sectors parameterize heterogeneous firm and worker agents whose monthly interactions on product and labour markets generate emergent macro-level dynamics. No sigmoid of time appears anywhere in the model code.

1 Introduction

The question of how artificial intelligence will transform employment has moved from speculative long-range forecasting (Frey and Osborne, 2017) to an urgent empirical and policy debate, as large language models demonstrably perform a substantial share of economically relevant tasks (Eloundou et al., 2024; Noy and Zhang, 2023; Brynjolfsson et al., 2025). The dominant analytical tradition treats this question at the aggregate level: task-based production frameworks decompose occupations into automatable and complementary components (Autor et al., 2003; Acemoglu and Restrepo, 2018, 2019), and scenario analyses trace out macroeconomic paths under assumptions about the speed and breadth of automation (Korinek and Suh, 2024; Korinek, 2024). These approaches are powerful, but they share a structural limitation: the dynamics of interest—the timing of adoption take-off, the depth of transitional unemployment, the distribution of wage losses—are typically encoded directly in functional forms such as logistic diffusion curves and aggregate matching functions. The model then illustrates its assumptions rather than testing whether the posited dynamics can arise from plausible micro-behaviour.

This paper takes the complementary, generative route (Epstein and Axtell, 1996; Tesfatsion, 2006). We present the design and first results of an agent-based model (ABM) in which the units of analysis are the actual decision-makers: firms that weigh the return on automating specific tasks against adjustment costs and organizational inertia, and workers who search for jobs, form reservation wages, and retrain toward occupations whose task content resembles what they already know. The model’s central discipline is that *nothing tips by construction*. Adoption accelerates only if falling AI prices, advancing capability, heterogeneous adoption hurdles, and imitation externalities jointly produce acceleration; unemployment spikes only if displacement outruns a decentralized matching process; inequality rises only if automation reprices some workers’ marginal products relative to others.

The contribution is threefold. First, we describe a compact, fully reproducible model architecture—roughly 930 lines of tested Python built on the Mesa 3 framework (Kazil et al., 2020; ter Hoeven et al., 2025)—that connects the task-exposure measurement literature (Eloundou et al., 2024; Felten et al., 2021; Webb, 2020; Handa et al., 2025) to search-and-matching labour-market dynamics (Mortensen and Pissarides, 1994; Pissarides, 2000) inside a single computational laboratory. Second, we introduce a calibration-gate methodology: the no-AI baseline is tuned to empirical labour-flow anchors and then *locked as an automated test suite* that every subsequent mechanism must keep green, which guards against the parameter drift that afflicts macro ABMs (Fagiolo and Roventini, 2017). Third, we report emergent outcomes with direct policy relevance, including a result that aggregate treatments cannot even express: the transitional-distress effects of unemployment benefits and employment protection are seed- and specification-dependent, with medians, dispersions, and even signs that shift across replications and model revisions, while only the efficiency of reallocation itself is consistently beneficial.

2 Related work

Task-based models of automation originate with Autor et al. (2003) and were generalized by Acemoglu and Restrepo (2018, 2019), who emphasize the race between displacement of existing tasks and reinstatement of new ones. Autor (2015) stresses the complementarity channel through which partial automation raises the productivity of remaining human tasks; our augmentation mechanism implements exactly this logic at the level of individual firm–occupation pairs. Occupation-level exposure measurement has produced a family of scores based on capability mappings (Felten et al., 2021; Webb, 2020), culminating in the E0/E1/E2 classification of Eloundou et al. (2024), which we adopt as the schema for our task data, and in observational usage evidence from deployed assistants (Handa et al., 2025). Micro-econometric estimates of AI’s productivity effects (Noy and Zhang, 2023; Brynjolfsson et al., 2025) and labour-demand effects (Hampole et al., 2025) provide the empirical magnitudes toward which the model’s placeholder parameters are oriented.

On the modelling side, our labour market descends from equilibrium search theory (Mortensen

and Pissarides, 1994; Pissarides, 2000), but replaces the aggregate matching function with an explicit application-and-screening protocol, so that matching efficiency is a mechanical parameter while the Beveridge relationship (Blanchard and Diamond, 1989) is an output. The macro-ABM tradition—the Keynes-meets-Schumpeter family (Dosi et al., 2010, 2018) and Eurace@Unibi (Dawid et al., 2019)—demonstrates that decentralized economies with boundedly rational firms reproduce rich labour-market regularities; we differ in focusing the entire model on a single technology shock with task-level resolution, at the cost of abstracting from money, credit, and aggregate demand feedbacks. Firm-size heterogeneity follows the power-law evidence of Axtell (2001). Our documentation approach follows the spirit of the ODD protocol (Grimm et al., 2020), and the implementation uses Mesa 3 (ter Hoeven et al., 2025) with a Solara-based visual front end (Kazil et al., 2020).

3 Model design

3.1 Entities and empirical grounding

The economy contains $N = 5,000$ workers (2,000 in interactive use) and approximately 400 firms distributed over the eight sectors of the model’s predecessor. Three bundled, schema-validated data files ground the agents. A task file defines 24 generalized work activities, each carrying an exposure class following Eloundou et al. (2024)—E0 (not exposed, e.g. physical care), E1 (exposed with LLM access), E2 (exposed with LLM plus scaffolding software)—together with a *difficulty* threshold (the capability level at which AI output quality becomes acceptable) and an *augmentation* coefficient (the productivity boost to a human performing the task with AI assistance). An occupation file defines 22 SOC major groups as weight vectors over tasks, with base wages and employment shares oriented to BLS magnitudes. A sector file expresses each sector as an occupation mix with an iso-elastic demand elasticity and a truncated-Pareto firm-size distribution (Axtell, 2001). All numerical values are transparently labelled placeholders pending curation (Section 7); every record carries a source field.

A key design decision is that **occupational distance is derived, not assumed**: the distance between occupations o and o' is one minus the cosine similarity of their task-weight vectors. Retraining feasibility, cross-occupation applications, and skill transferability all read from this geometry, so improving the task data automatically improves the mobility structure.

3.2 Decision rules and the monthly cycle

Each tick (one month) executes ten steps. (1) An AI capability frontier C_t follows a logistic path toward a ceiling with seeded AR(1) shocks, and the per-task automation quality is $q_k = \max\{0, (C_t^{\text{eff}} - d_k)/(1 - d_k)\}$, where d_k is task difficulty and effective capability is discounted for E1 tasks; the AI price p_t^{AI} declines exponentially. Capability and cost are deliberately separate levers, following the scenario axes of Korinek and Suh (2024). (2) Firms evaluate a small random sample of their tasks each month (bounded rationality) and adopt task k with probability $\Lambda(\rho_k + \iota \cdot A_s - h_f)$, where Λ is a logistic in the decision signal, $\rho_k = (w_f - p_t^{\text{AI}}/q_k - \kappa)/w_f$ is the ROI against the firm’s mean wage w_f with amortized adjustment cost κ , A_s is the sector’s current adoption share (imitation), and h_f is a firm-specific hurdle drawn from a normal distribution capturing management quality and organizational conservatism. Adoption is absorbing. (3) Automation and augmentation reduce unit costs; 60% passes through to prices; iso-elastic sector demand expands (the rebound is endogenous), and a size-weighted logit over prices splits a conserved sector volume across firms, so entrants dilute incumbents rather than adding demand. (4) Firms recompute per-occupation headcount targets as $\text{base} \times \text{demand factor} \times (1 - a_{fo})/(1 + g_{fo})$, where a_{fo} and g_{fo} are the automated share and augmentation boost of occupation o ’s tasks at firm f ; surplus workers are shed gradually at a rate damped by firing costs, lowest tenure first. (5) Employed workers separate exogenously at 1.2% per month. (6) Vacancies are posted for deficits, with offers escalating 2% per month unfilled. (7) Searchers send five applications, weighted toward occupations near their own; workers unemployed beyond a threshold search the full occupation space, carrying a skill discount δ^{dist} that

Table 1: No-AI baseline against calibration anchors (seed 42, $N = 2,000$, 20 years, burn-in 24 months). Anchor bands are enforced as automated tests.

Moment	Anchor band	Model	Empirical referent
Mean unemployment rate	3–8%	3.5%	U.S. postwar average $\approx 5\%$
Monthly job-finding rate	10–45%	25.2%	$\approx 25\text{--}30\%$ (Pissarides, 2000)
Monthly separation rate	0.6–2.5%	1.2%	$\approx 1.5\text{--}2\%$
Wage Gini (final)	0.20–0.45	0.30	earnings Gini ≈ 0.35
Employment drift (20y)	$ \Delta < 3$ pp	+1.4 pp	stationarity

the retraining subsidy attenuates. Firms rank applicants by effective skill, and a screening-efficiency coin gates completion. (8) Hire wages split marginal product and reservation wage at Nash weight β ; reservation wages decay 1% per month of unemployment toward a benefit floor; 24-month spells transition to discouragement, from which workers re-enter at 3% per month. (9) Employed wages drift toward marginal products that automation lowers and augmentation raises—the wage-scarring channel. (10) Firms with persistently deficient demand exit, displacing their workforce; exits are replaced by entrants with *lower* adoption hurdles (a vintage effect), and booming sectors attract additional entry.

3.3 Calibration gates and reproducibility

Calibration proceeds in a strict order. The no-AI baseline (capability fixed at zero) was first tuned to four empirical anchors—mean unemployment of roughly 5%, monthly job-finding near 25%, monthly separations near 1.5%, and a wage Gini near 0.3—and these bands were then frozen into the automated test suite together with property-based invariants (worker conservation, employer–employee link integrity, bit-identical series under identical seeds). Every AI mechanism added later had to keep this gate green. The discipline paid off concretely: during development the gate caught an over-efficient matching market, a market-share normalization that let entrants create demand from nothing, and a time-zero inconsistency in which initial capability granted an instantaneous augmentation windfall that triggered a spurious layoff wave. The final suite contains 60 tests with 96% line coverage of the engine. A single master seed spawns independent `numpy` streams per subsystem and derived sub-seeds per Monte Carlo run, so any figure in this paper can be regenerated exactly.

4 Results

4.1 Baseline validation

Table 1 reports the no-AI baseline against its calibration anchors over a twenty-year horizon. The stationary economy holds unemployment in a narrow band around 3–5% with realistic gross flows and no secular employment drift, confirming that subsequent AI-driven dynamics are attributable to the AI mechanisms rather than to baseline disequilibrium.

4.2 Scenario dynamics: emergent S-curves and displacement with rebound

Figure 2 contrasts the baseline preset (moderate capability growth) with a fast-takeoff scenario (monthly capability growth 8%, AI cost decline 3%, strong imitation, low hurdles) and a high-friction scenario. Firm task adoption traces a canonical S-curve—a quiet build-up while capability clears few task difficulties, then an abrupt cascade as ROI, imitation, and entrant vintages compound, then saturation near the exposed task mass—despite no diffusion curve being specified anywhere. The automated test suite asserts this property: adoption at month twelve must remain below a quarter of its final value, and the fastest adoption growth must occur after the first year. The three scenarios reach nearly identical long-run output gains of roughly 80%; what varies is *timing*. The cascade arrives around year two under fast takeoff versus year four in the baseline, and the labour market pays for speed: an early distress spike near 20% of the workforce accompanies each cascade,

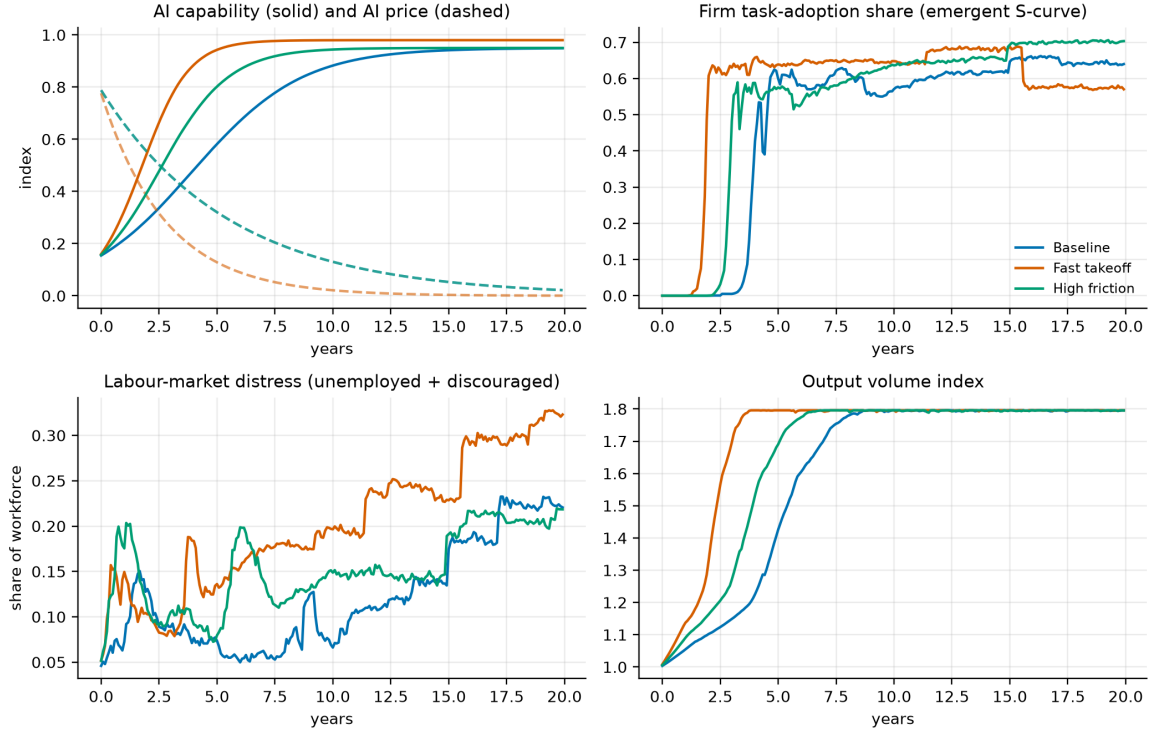


Figure 2: Scenario comparison (seed 42, $N = 2,000$, 20 years). Top left: capability frontier (solid) and AI price (dashed). Top right: emergent adoption S-curves. Bottom: labour-market distress and output. Nothing in the code imposes the S-shape.

and under sustained automation distress ratchets upward in steps—each step a wave of firm exits—reaching roughly a third of the workforce by year twenty under fast takeoff. Productivity gains and severe, persistent transitional pain coexist in the same runs, the central tension emphasized by Korinek (2024).

4.3 Uncertainty quantification

Because tipping dynamics are nonlinear, single runs can mislead. Figure 3 shows percentile bands over 24 seeded replications of the fast-takeoff scenario. The *timing* of the adoption cascade is tightly determined—it is driven by the capability process—but uncertainty grows along the path: the median unemployment peak of roughly 16% at the cascade spans about 14–20% across seeds, and by year twenty the p5–p95 band covers 11–23% unemployment and 0.47–0.71 adoption, reflecting the path dependence of matching congestion and firm exit cascades. Interquartile bands rather than single trajectories are therefore the appropriate reporting unit for policy claims, and the application exposes them directly.

4.4 Inequality and reallocation stress

The left panel of Figure 4 shows the wage-percentile fan under fast takeoff, and the result is starker than a widening of dispersion: the median wage roughly halves over the transition as automation reprices the marginal product of exposed task mass across nearly all occupations, and displaced workers re-hired after reservation-wage decay lock in persistent scars. Notably, the *within-labour* Gini drifts slightly *down*—the wage distribution compresses toward the benefit floor even as its level collapses. The model thus distinguishes two inequality concepts that aggregate discussion often conflates: dispersion among workers falls while the implied labour share of surplus (output nearly doubles as wages halve) shifts dramatically toward firms and, in a richer model, capital owners. The right panel traces the Beveridge curve. The no-AI baseline occupies a tight low-unemployment locus, while the fast-takeoff economy sweeps through a two-phase loop: a vacancy-rich reallocation phase during the adoption cascade, with vacancy rates transiently reaching tens of percent as restructuring

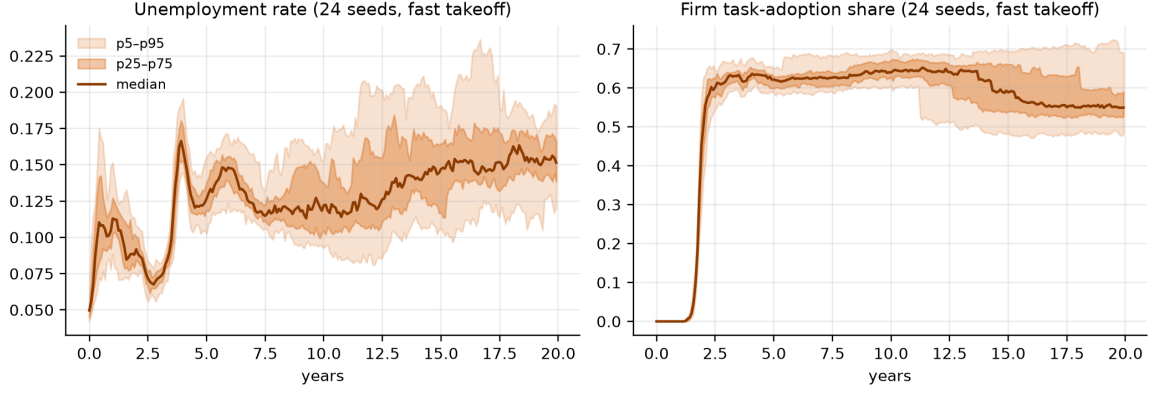


Figure 3: Monte Carlo uncertainty bands over 24 seeded replications (fast takeoff, $N = 1,500$). Shaded regions are p5–p95 and p25–p75; lines are medians.

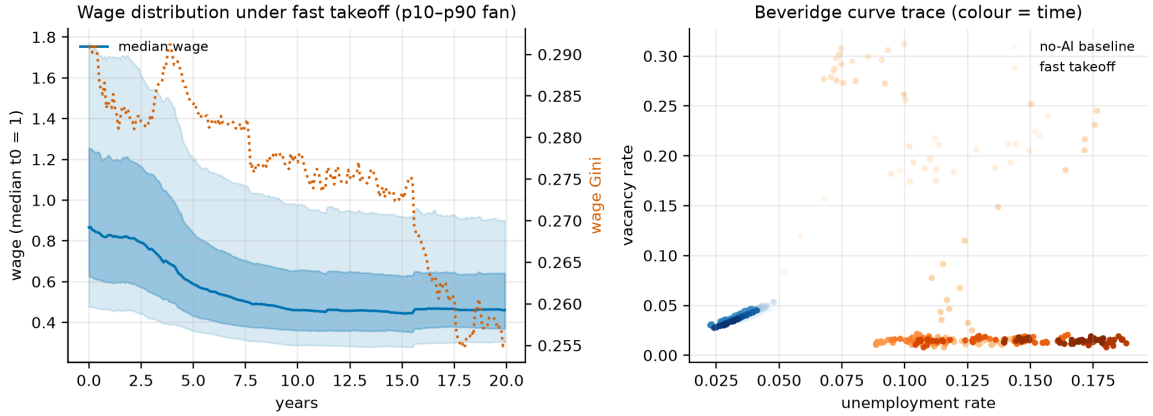


Figure 4: Left: wage-percentile fan (p10–p90) and Gini under fast takeoff. Right: Beveridge-curve traces for the no-AI baseline and fast takeoff; colour encodes time.

firms and entrants repost positions faster than the matching process can fill them, followed by a slack regime far to the south-east where low vacancy rates coexist with 10–19% unemployment. This is the classic signature of reallocation-driven mismatch (Blanchard and Diamond, 1989), generated here by the distance structure of the occupation space rather than by an assumed mismatch index.

4.5 Policy experiments: non-robust levers

Figure 5 reports peak distress across five seeds for three two-level experiments under fast takeoff, and the headline finding is *non-robustness* rather than any single sign. Raising the benefit floor from 0.25 to 0.65 of the median wage lowers median peak distress from 0.293 to 0.228—the income floor stabilizes re-employment wages and keeps discouraged workers attached—yet two of five seeds worsen, because the counteracting reservation-wage channel (generous benefits hold reservation wages up exactly when re-employment requires accepting lower-paid work, the classic moral-hazard margin of search theory, Pissarides, 2000) dominates on some tipping paths. Strikingly, an earlier development revision of the product-market block produced the *opposite* median ordering with the same mechanisms present; the sign of the benefit effect is a property of the specification, not of the model family. Raising firing costs from 0.05 to 0.60 leaves the median essentially unchanged (0.240 vs. 0.234) while flipping sign seed-by-seed: employment protection spreads displacement over time on some paths but delays it into a synchronized, congested layoff wave on others. Only matching efficiency—the throughput of the reallocation process itself—delivers a consistent median improvement (0.226 fluid vs. 0.266 frictional), and it is the only monotonicity the test suite is allowed to enforce; even here individual seeds deviate, so medians over replications, not runs, are the unit of inference. We stress the epistemic status of these findings: they are properties of a placeholder-calibrated model, valuable as hypothesis generators and as a demonstration that an ABM can express—and expose—trade-offs

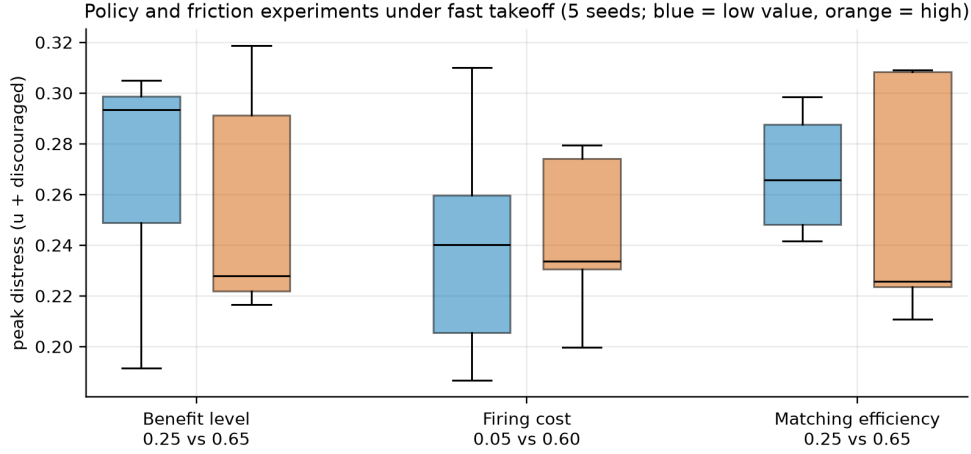


Figure 5: Peak distress under fast takeoff across five seeds for three two-level experiments (blue = low value, orange = high value). Benefits help in the median but not in every seed; firing costs are a wash with seed-level sign flips; matching efficiency delivers the only consistent median improvement.

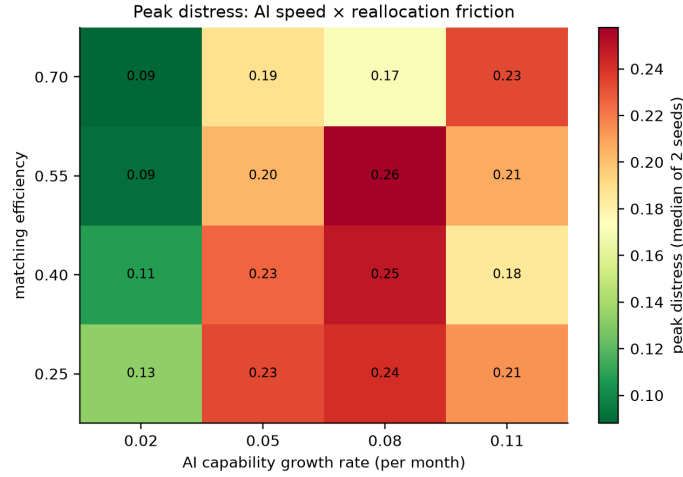


Figure 6: Peak distress over the AI-speed × reallocation-friction grid (median of 2 seeds per cell, $N = 800$, 12 years).

and specification risk that aggregate models bury in functional forms. Their ambiguity is consistent with the mixed empirical evidence on labour-market programmes (Card et al., 2018).

Figure 6 maps peak distress over the joint space of AI capability growth and matching efficiency. The dominant feature is a threshold in speed: distress roughly doubles between slow takeoff (2% monthly capability growth, peak distress 0.09–0.13) and moderate takeoff, and then approximately plateaus—once the adoption cascade outruns the market’s monthly reallocation capacity, additional speed adds little further damage. Higher matching efficiency lowers distress across most of the grid, most cleanly at slow and moderate speeds. At the fastest growth rates the two-seed cell medians are visibly noisy, echoing the seed sensitivity of Section 4.5 and motivating the denser replication proposed in Section 7.

5 The interactive application

The model ships as a two-page web application (Figure 7). The *Live model* page couples SolaraViz play/step/reset controls and parameter sliders to an agent-space view in which workers appear as dots coloured by employment state, clustered around employer firms drawn as squares sized by headcount and coloured by AI adoption share; unemployed workers drift to a central ring. Time-series panels, a wage-percentile fan, a Beveridge trace, an occupation-flow matrix, and CSV export update each tick. The *Research* page runs seeded Monte Carlo batches with live progress, overlays two experiments’

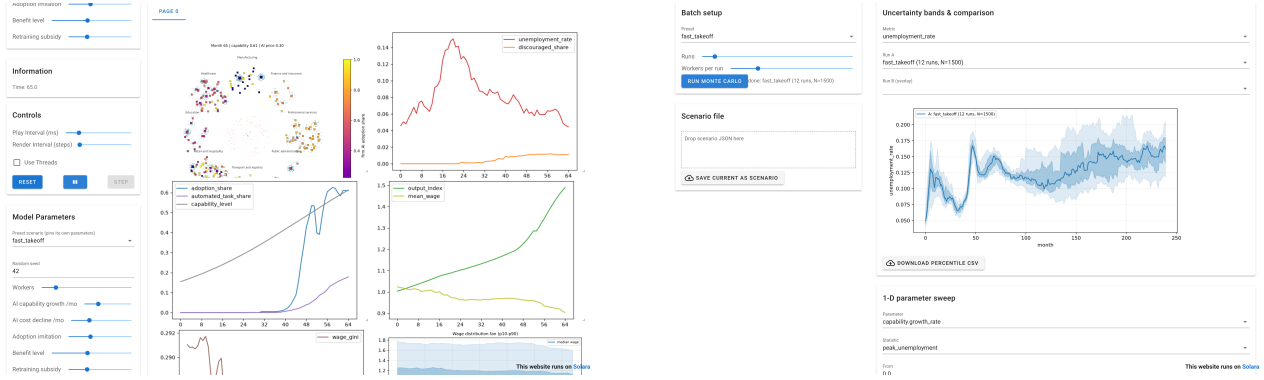


Figure 7: The interactive application. Left: Live model page with agent-space view and time-series panels. Right: Research page with Monte Carlo bands and parameter sweeps.

percentile bands, saves and loads versioned scenario files, and executes one-dimensional parameter sweeps. The visual mode has proved valuable for hypothesis generation—adoption cascades are literally visible as colour fronts sweeping through sector clusters—while the batch mode supplies the statistical discipline that single animated runs lack.

6 Discussion and limitations

The exercise supports three methodological claims. First, the generative standard is attainable at modest scale: a model small enough to audit line-by-line reproduces S-curve diffusion, displacement-with-rebound, wage scarring, and Beveridge shifts from first principles. Second, calibration gates—empirical anchor bands frozen as tests before new mechanisms land—convert the folklore of ABM discipline (Fagiolo and Roventini, 2017; Grimm et al., 2020) into an executable practice; each of the three serious specification errors we made was caught by the gate, not by inspection. Third, micro-founded policy analysis surfaces ambiguities that matter: the seed- and specification-sensitivity of the benefit and firing-cost effects in Section 4.5 would be invisible in a model whose transition path is a dial—and a modeller who stopped at a single seed, or a single specification, would have reported a confident sign that does not survive replication.

The sensitivities carry practical content of their own, because they tell decision-makers *where* confidence is and is not available. For policymakers, three implications follow. Because the levers with the greatest political salience—benefit generosity and employment protection—are exactly those whose effects proved fragile, transition policy is better designed as an adaptive *portfolio* (income support bundled with retraining subsidies and matching investment, with pre-agreed adjustment triggers) than as a commitment to any single instrument with an assumed sign. Guaranteed resources belong instead with the one robustly signed lever, reallocation capacity: job-search assistance, credential portability, and transparent skill taxonomies—the intervention class for which empirical evidence is also least ambiguous (Card et al., 2018), and whose value in the sweep is concentrated precisely in fast-takeoff regions where braking adoption buys little beyond the congestion threshold. And because the cascade’s *timing* is tightly determined even when its consequences are not (Figure 3), monitoring dominates forecasting: adoption acceleration, vacancy-to-searcher ratios, and Beveridge-curve displacement are observable precursors that can arm state-contingent responses. A final measurement warning follows from the inequality results (Figure 4): a regulator watching wage dispersion alone would read the transition as equalizing while wage *levels* collapse; the labour share, not the within-labour Gini, is the statistic to track.

For managers the model reads as a survival analysis. The firm trait that most determines exit during the transition is the idiosyncratic adoption hurdle—organizational conservatism and evaluation capability—whose practical counterpart is a standing discipline of task-level ROI review rather than episodic “AI initiatives”; and because complementarity operates *within* occupations, that review must decompose jobs into tasks, automating selectively and redesigning roles around the residual and augmented bundle (Autor, 2015). Imitation dynamics imply that most firms restructure exactly when their sector does, entering the labour market during the vacancy-rich mismatch phase of the

Beveridge loop; early movers hire and retrain at pre-cascade conditions, while mid-cascade restruc-
turers should expect long time-to-fill and treat retention of task-adjacent staff as cheaper than search.
The task-distance geometry doubles as an internal redeployment map—mobility succeeds between
task-adjacent roles and fails across distant ones—and the seed-level variance of Figure 3 argues
for staged, option-like AI investment over big-bang commitments, for the same reason the policy
conclusion favours triggers over fixed rules. Institutions that build monitoring, adaptability, and
reallocation capacity are robust to the model being wrong in either direction; institutions optimized
against one assumed transition path are fragile in every direction but one.

The limitations are substantial and worth stating plainly. All task, occupation, and sector
parameters are placeholders; magnitudes (though not, we believe, mechanisms) should be treated
as illustrative. The model omits robotics (E0 tasks never automate), aggregate-demand feedback
from unemployment to consumption, capital markets, firm-level AI implementation failures, within-
occupation skill heterogeneity beyond a scalar, and any geography. Discouragement and re-entry are
stylized. Firms do not strategically anticipate AI progress; adoption is myopic. Finally, a 24-task,
22-occupation grid is coarse relative to the thousands of O*NET work activities.

7 Future research and refinement

Five directions follow directly. *First, empirical curation.* Replace placeholder exposure, difficulty,
and augmentation values with the published task-level estimates of Eloundou et al. (2024) and
observed usage intensities from Handa et al. (2025); weight occupations from O*NET work activi-
ties and BLS OES employment; validate the implied occupation-distance matrix against observed
occupational-transition matrices. *Second, mechanism extensions.* Add a robotics channel so E0 tasks
automate on a slower, capital-intensive schedule; endogenize task creation (Acemoglu and Restrepo,
2019) so reinstatement competes with displacement; introduce aggregate-demand feedback and firm
balance sheets in the spirit of Dosi et al. (2010); let firms form expectations over the capability path.
Third, deeper validation. Confront the model with the emerging quasi-experimental evidence on AI
exposure and hiring (Hampole et al., 2025), and test whether simulated occupational wage responses
match early observational estimates; historical automation episodes (spreadsheets, industrial robots)
offer out-of-sample transitions for back-casting. *Fourth, systematic exploration.* The seed-level vari-
ance visible in Figure 3 calls for global sensitivity analysis (e.g. Sobol indices over the full parameter
space) and for scaling tests confirming that results at $N = 5,000$ persist at $N = 50,000$. *Fifth,
richer policy design.* The non-robust single-lever results invite optimal-policy search over *combina-
tions*—benefit generosity bundled with retraining subsidies and matching-technology investment—
and evaluation of timing rules that condition policy on observed adoption acceleration, a quantity
the model’s tipping diagnostics already measure.

More broadly, the calibration-gate pattern and the derived-mobility-structure pattern seem trans-
ferable to other technology-transition ABMs, and packaging them as reusable methodology is itself
a research contribution we intend to pursue.

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